

AVIGNON PROVENCE AP, France

WMO: 075630

Lat: 43.9114N

Lon: 4.9025E

Elev: 48

StdP: 100.75

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.7	(c) -2.2	(d) -9.7	(e) 1.7	(f) 0.9	(g) -7.2	(h) 2.1	(i) 1.7	(j) 17.0	(k) 5.3	(l) 15.1	(m) 5.8	(n) 2.9	(o) 350	(p) 0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 34.8	(d) 21.5	(e) 33.3	(f) 20.8	(g) 31.9	(h) 20.4	(i) 23.0	(j) 31.6	(k) 22.2	(l) 30.3	(m) 21.4	(n) 29.2	(o) 4.0	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.1	25.9	19.6	14.4	25.4	18.8	13.7	24.9	68.1	31.4	65.0	30.4	62.3	29.1	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.7	(b) 12.0	(c) 10.5	(e) -6.1	(f) 37.6	(g) 1.8	(h) 1.8	(i) -7.3	(j) 38.9	(k) -8.4	(l) 40.0	(m) -9.4	(n) 41.0	(o) -10.7	(p) 42.3		
(a) 13.7	(b) 12.0	(c) 10.5	(e) -6.7	(f) 24.5	(g) 2.0	(h) 0.8	(i) -8.2	(j) 25.1	(k) -9.4	(l) 25.5	(m) -10.5	(n) 26.0	(o) -12.0	(p) 26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	6.1	6.8	10.4	14.0	17.5	22.2	25.0	24.0	20.2	15.9	10.3	6.6		
	DBStd	7.33	3.37	3.51	3.17	2.85	2.94	2.96	2.37	2.43	2.77	3.31	3.75	3.52		
	HDD10.0	418	127	96	33	3	0	0	0	0	0	2	41	116		
	HDD18.3	1837	378	322	246	132	51	3	0	0	13	87	241	364		
	CDD10.0	2232	7	7	46	123	232	366	467	434	306	186	50	10		
	CDD18.3	611	0	0	0	2	24	119	208	177	68	12	0	0		
	CDH23.3	6162	0	0	3	49	262	1238	2233	1739	574	63	0	0		
	CDH26.7	2581	0	0	0	3	56	506	1072	775	166	4	0	0		
(14) Wind	WSAvg	4.0	4.3	4.8	4.3	4.0	4.2	3.8	3.9	3.8	3.6	3.8	4.0	3.8		
(15) Precipitation	PrecAvg	666	54	36	34	62	49	38	28	35	102	90	91	47		
	PrecMax	991	152	127	88	158	123	103	106	98	256	165	252	186		
	PrecMin	417	0	0	0	6	4	2	1	8	29	30	17	0		
	PrecStd	158	40	32	27	36	30	27	27	24	59	40	60	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	18.8	23.4	27.0	30.6	36.0	36.4	36.8	32.8	26.9	20.6	16.9		
		MCWB	12.1	11.7	13.0	16.7	18.8	22.3	22.1	21.9	20.3	18.4	15.0	13.3		
	2%	DB	15.1	16.4	20.4	24.8	28.1	33.4	34.8	34.6	30.4	25.0	19.2	15.5		
		MCWB	10.7	10.5	12.7	15.9	17.7	21.0	21.7	21.3	19.6	18.2	14.9	12.1		
	5%	DB	13.6	14.5	18.7	22.8	26.2	31.7	33.6	32.8	28.6	23.6	17.8	14.1		
		MCWB	10.1	9.6	11.8	14.8	16.9	20.4	20.8	20.9	19.0	17.5	14.1	11.2		
	10%	DB	12.1	13.0	16.9	20.8	24.4	29.8	32.2	31.0	26.8	22.0	16.3	12.6		
		MCWB	8.9	8.9	11.0	13.7	16.2	19.7	20.5	20.1	18.5	16.6	13.2	9.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	12.3	14.2	17.7	19.8	23.8	24.0	23.6	22.2	20.1	16.3	14.0		
		MCDB	16.2	17.1	19.9	25.2	27.5	34.4	32.6	32.8	28.3	24.7	19.0	16.2		
	2%	WB	11.5	11.2	13.3	16.4	18.7	22.2	23.1	22.8	21.1	19.1	15.3	12.6		
		MCDB	14.5	14.9	18.9	23.5	26.0	31.0	31.6	31.2	27.1	23.6	18.2	14.9		
	5%	WB	10.4	10.3	12.5	15.2	17.8	21.2	22.3	21.9	20.2	18.2	14.5	11.5		
		MCDB	13.2	13.8	17.6	21.5	24.4	29.5	30.5	29.9	26.0	22.5	17.3	13.7		
	10%	WB	9.2	9.2	11.6	14.2	16.9	20.3	21.5	21.1	19.3	17.1	13.6	10.2		
		MCDB	11.7	12.5	16.0	19.7	23.0	28.1	29.5	28.6	25.0	21.0	16.0	12.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.6	11.2	11.8	11.8	12.7	13.3	13.1	12.0	10.3	8.8	8.6		
		MCDBR	10.7	13.0	14.5	15.5	14.9	15.9	15.7	16.3	15.1	12.2	10.6	10.1		
		MCWBR	7.1	8.1	8.0	8.0	6.7	6.1	5.4	5.9	6.1	6.4	6.9	7.1		
	5% WB	MCDBR	9.6	11.7	12.8	14.3	13.7	15.0	14.5	14.7	12.9	11.0	9.5	8.8		
		MCWBR	6.7	7.8	7.4	7.6	6.7	6.2	5.7	6.2	5.9	6.1	6.5	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.343	0.382	0.398	0.407	0.407	0.399	0.397	0.386	0.388	0.350	0.320		
	taud		2.454	2.394	2.299	2.281	2.303	2.332	2.353	2.364	2.385	2.372	2.430	2.472		
	Ebn at Noon		800	844	850	866	866	865	867	857	838	779	757	767		
	Edh at Noon		73	92	115	126	127	124	120	115	104	94	75	66		
(44) All-Sky Solar Radiation	RadAvg		1.70	2.66	3.96	5.15	6.20	7.20	7.20	6.17	4.70	2.95	1.87	1.48		
	RadStd		0.18	0.26	0.33	0.41	0.45	0.33	0.33	0.23	0.28	0.28	0.23	0.16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page